

ORIGINAL RESEARCH



Extra-Carotid Arterial Webs: A Single-Center Series and Systematic Review

Eric A. Grin¹ ID, BS; Evan G. Stein² ID, MD, PhD; Lara Balick, MPH; Austin Chen, MD; Scott E. Kasner³ ID, MD; Sara K. Rostanski⁴ ID, MD; Erez Nossek, MD; Michela Rosso⁵ ID, MD

BACKGROUND: Carotid webs are increasingly recognized as a cause of ischemic stroke, but less is known about morphologically similar lesions in other arteries. We present the first study characterizing the clinical and radiographic features of extra-carotid arterial webs through a single-center case series and systematic review.

METHODS: Patients with possible extra-carotid webs were identified from 2017 to 2025 using a natural language processing search of radiology reports at our institution. Candidate cases underwent imaging review with multiplanar and 3-dimensional reconstructions to distinguish webs from fenestrations, vessel tortuosity, dissection, or atherosclerotic plaque. In parallel, we performed a Preferred Reporting Items for Systematic Reviews and Meta-Analyses–compliant systematic review (Web of Science and PubMed, inception to September 2025) to identify published cases of extra-carotid web. Data on demographics, vascular location, imaging modality, clinical presentation, treatment, and outcomes were extracted and synthesized descriptively.

RESULTS: Four extra-carotid webs were identified at our institution (3 basilar and 1 subclavian). None were associated with stroke, and all patients remained stable on conservative management during 9 months to 4 years of follow-up. Across 16 published studies, 22 additional extra-carotid webs were identified, yielding 26 patients in aggregate (mean age, 52.7 years; 61.5% male). The vertebral artery was the most common site (53.8%), followed by the basilar (30.8%) and subclavian (15.4%) arteries. Ten patients with vertebral or basilar web presented with posterior circulation ischemic stroke, with recurrent events in 4 patients. Most patients were managed with antiplatelet therapy or observation, while 3 vertebral webs with recurrent stroke were successfully treated with stenting.

CONCLUSIONS: Extra-carotid webs share morphological and clinical features with carotid webs and may represent a unified disease spectrum of cervical artery webs. Vertebral and basilar webs, though rare, may be an underrecognized source of posterior circulation stroke. Recognition of these lesions may broaden the differential for cryptogenic stroke though the lack of histopathologic visualization remains a critical limitation of our study.

Key Words: carotid webs ■ ischemic stroke ■ systematic review

While historically underappreciated, carotid webs are now an increasingly recognized cause of ischemic stroke, particularly in young patients.^{1,2} On vascular imaging, carotid webs appear as thin, often shelf-like intraluminal projections at the carotid bulb. They disrupt laminar flow, creating regions of flow stasis that promote thrombosis and downstream embolization.^{2–4} Histologically, carotid webs show fibromuscular intimal thickening though the underlying cause of these lesions remains uncertain.⁵

Although carotid webs are now established as a significant cause of ischemic stroke, much less is known about morphologically similar lesions in other arteries. The first description of an extra-carotid web was published in 2000 when Green and Caplan⁶ reported a basilar artery web in an 18-year-old boy presenting with pontine stroke. More recently, additional cases have been sporadically reported in the vertebral, basilar, and subclavian arteries, but systematic evaluation is lacking.^{7–10}

Correspondence to: Michela Rosso, MD, Department of Neurology, NYU Grossman School of Medicine, 550 1st Ave, New York, NY 10016. Email michela@mrosso.us
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Nonstandard Abbreviations and Acronyms	
CTA	computed tomography angiography
DSA	digital subtraction angiography
FMD	fibromuscular dysplasia
LDL	low-density lipoprotein

CLINICAL PERSPECTIVE

What Is New?

- This research provides the first systematic description to characterize extra-carotid arterial webs (found in vertebral, basilar, and subclavian arteries), combining a single-center series (n=4) with a systematic review (n=22).

What Are the Clinical Implications?

- Arterial webs share radiographic morphology regardless of location and should be recognized as a potential, underrecognized source of cryptogenic stroke.
- Recognizing cervical artery webs as a unified entity may improve diagnostic precision, align management with recurrence risk, and guide further research in clarifying these lesions' natural history, optimal management strategies, and underlying cause.

Given their morphological similarity to carotid webs, extra-carotid lesions could plausibly represent part of the same disease spectrum. Recognizing arterial webs as a broader entity is important both for advancing understanding of their cause and pathogenesis and for identifying vertebral and basilar webs as underrecognized causes of thromboembolic disease and, specifically, posterior circulation stroke.

In this study, we report a single-center series of 4 extra-carotid webs in addition to a systematic review of the literature, synthesizing the clinical and radiographic features of extra-carotid webs and identifying areas for future investigation.

METHODS

Data Availability Statement

The data supporting this study's findings are available from the corresponding author upon reasonable request.

Case Series

Patient Identification

Patients with extra-carotid webs were identified using a previously described natural language processing approach applied to the Epic electronic health record.¹¹ Briefly, this approach leverages a large language model pretrained on unstructured clinical

notes to directly extract relevant information from the Epic electronic health record without requiring manual preprocessing.

The search included all radiology reports of computed tomography angiography (CTA), magnetic resonance angiography, or digital subtraction angiography (DSA) performed at our institutions (NYU Langone Health Medical Center, Manhattan, and NYU Langone Hospital, Brooklyn) between July 2017 and April 2025, as described in our prior work.⁴ Reports were queried for the plain-language terms web, shelf, or protrusion, yielding 1026 candidate records. Institutional review board approval with waiver of consent was obtained.

Screening and Selection

Imaging reports were manually reviewed to exclude cases in which the terms web, shelf, or protrusion referred to unrelated devices (eg, Woven EndoBridge) were used in incidental contexts, described carotid webs only, or mentioned arterial web only as a highly unlikely differential (Figure S1). Twenty candidate extra-carotid cases were identified and independently reviewed by a board-certified neurovascular radiologist with expertise in cerebrovascular imaging. The reviewer was blinded to clinical data, including patient demographics, presentation, and outcomes, to minimize assessment bias.

Imaging review incorporated multiplanar reconstructions, curved planar reformatting, and 3-dimensional reconstructions when available. Extra-carotid webs were defined radiographically as thin, linear intraluminal filling defects with smooth borders, lacking calcification, and morphologically distinct from atherosclerotic plaque, fenestration, or focal dissection. Candidate lesions were systematically differentiated from potential mimics.

Fenestrations were excluded by confirming continuity with both proximal and distal true lumens on multiplanar and 3-dimensional reconstructions. Dissection was excluded when no intimal flap, double lumen, tapered stenosis, or mural hematoma was identified. Arterial webs also present as focal, fixed endoluminal abnormalities rather than the longitudinal wall process typically seen in acute or healed dissection. Vessel tortuosity or kinking was excluded by assessing the vessel course in multiple planes to ensure the shelf-like appearance was not projectional. Finally, atherosclerotic plaque was considered a more likely diagnosis than arterial web when there was circumferential wall thickening, calcification, or irregular surface morphology, especially in the context of calcified and noncalcified atheroma in adjacent vessels. Small plaques, even when nonstenotic, more commonly demonstrate a broader base or smoother contours, whereas arterial webs project more sharply into the lumen with an abrupt margin and narrow base.

When serial imaging was available, lesions were confirmed to be morphologically stable and positionally fixed across studies. Lesions were included only when an arterial web was judged the most likely diagnosis after excluding competing differentials. After review, 4 patients, 3 with basilar webs and 1 with a subclavian web, met the inclusion criteria.

Systematic Review

Search Query

This systematic review was performed according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses

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guidelines. Data are available from the corresponding author upon reasonable request.

The Web of Science and PubMed were searched from inception to September 6, 2025, to identify all reports of arterial webs. To ensure comprehensive capture of the literature, the reference lists of all articles identified during full-text review were manually screened. No limitations were placed on language, publication type, or publication date. Controlled vocabulary and natural language searching were used for the following search terms:

Web of Science

ALL=(“carotid web” OR “carotid webs” OR “carotid artery web” OR “carotid bulb diaphragm” OR “intimal variant of fibromuscular dysplasia” OR “intimal fibromuscular dysplasia” OR “fibromuscular dysplasia, intimal” OR “intimal fibromuscular hyperplasia” OR “vascular web” OR “cervical web” OR “artery web” OR “basilar web” OR “subclavian web” OR “vertebral web” OR “atypical fibromuscular dysplasia”).

PubMed

“Carotid web”[tiab] OR “carotid webs”[tiab] OR “carotid artery web”[tiab] OR “carotid bulb diaphragm”[tiab] OR “intimal variant of fibromuscular dysplasia”[tiab] OR “intimal fibromuscular dysplasia”[tiab] OR “fibromuscular dysplasia, intimal”[tiab] OR “intimal fibromuscular hyperplasia”[tiab] OR “vascular web”[tiab] OR “cervical web”[tiab] OR “artery web”[tiab] OR “basilar web”[tiab] OR “subclavian web”[tiab] OR “vertebral web”[tiab] OR “atypical fibromuscular dysplasia”[tiab].

Eligibility

Studies were included if they reported at least 1 arterial web outside the carotid artery. Exclusion criteria were (1) carotid-only webs, (2) classic medial fibromuscular dysplasia (FMD) or renovascular disease, (3) nonarterial webs, and (4) animal studies or otherwise irrelevant studies. For the purposes of the systematic review, the diagnosis of arterial web was accepted as reported by the original authors; independent radiological readjudication of published images was not performed.

Data Abstraction and Collection

Data abstraction was performed independently by 2 authors. Any discrepancies were addressed through discussion and resolved by consensus. For studies reporting broader cohorts, only patients with extra-carotid webs were included. Variables recorded were age, sex, vessel involved, presence of cardiovascular risk factors (hypertension, hyperlipidemia, smoking history, diabetes, and coronary artery disease), symptomatic or asymptomatic clinical presentation (meaning association with an ipsilateral stroke or transient ischemic attack), imaging modality, treatment type, and months of follow-up.

Given the expected small sample size due to the rarity of this condition, only descriptive analyses were performed. Continuous variables were presented as mean±SD, or median with interquartile range, based on their distribution. Categorical

variables were reported as frequency (column percent). No imputation was performed for missing data. When data were missing, the denominator reflected the number of participants with available information.

RESULTS

Case Series

Case 1

A 76-year-old White man with well-controlled hyperlipidemia on rosuvastatin (LDL [low-density lipoprotein], 93 mg/dL) presented to an outside institution with acute-onset dizziness upon awakening. Neurological examination, vital signs, and gait were normal. He described the dizziness as not purely lightheadedness nor purely vertigo, with no clear triggers. He denied recent illness, chest pain, palpitations, or personal or family history of cardiovascular disease. The episode was attributed to a peripheral cause, and he was discharged. Shortly after, he presented to our institution for further evaluation.

Brain magnetic resonance imaging showed no acute or chronic infarction. Magnetic resonance angiography with time-of-flight sequencing demonstrated a basilar artery web, appearing as a smooth, linear, shelf-like endoluminal projection with mild luminal stenosis (Figure 1A through 1C). The lesion also enhanced on a postcontrast T1-weighted magnetization-prepared rapid gradient-echo sequence, and there was no calcification or atherosclerosis in intracranial or extracranial vessels.

The patient's dizziness resolved spontaneously over 3 days. He was prescribed aspirin, which he self-discontinued after 2 weeks, electing to take turmeric supplements instead. Over 4 years of follow-up, he has remained asymptomatic without recurrent neurological events or repeat vascular imaging, maintaining an active lifestyle.

Case 2

A 31-year-old White man with a history of patent foramen ovale–related ischemic left MCA territory stroke status post-closure (no residual deficits), maintained on clopidogrel, presented with a persistent right-sided headache associated with nausea and vomiting. His medical history was significant for migraines and former cigarette smoking (2.5 pack-years, quit 3 years prior). He described this episode as similar to prior migraines but more severe and unrelieved with acetaminophen. Neurological examination was nonfocal. Noncontrast head computed tomography was unremarkable.

CTA of the head and neck demonstrated a shelf-like projection nearly perpendicular to the basilar artery wall with only mild luminal stenosis. These findings were confirmed on a later time-of-flight magnetic resonance angiography (no contrast administered) and were consistent with a basilar artery web with only mild luminal stenosis (Figure 2A through 2C). There was no evidence

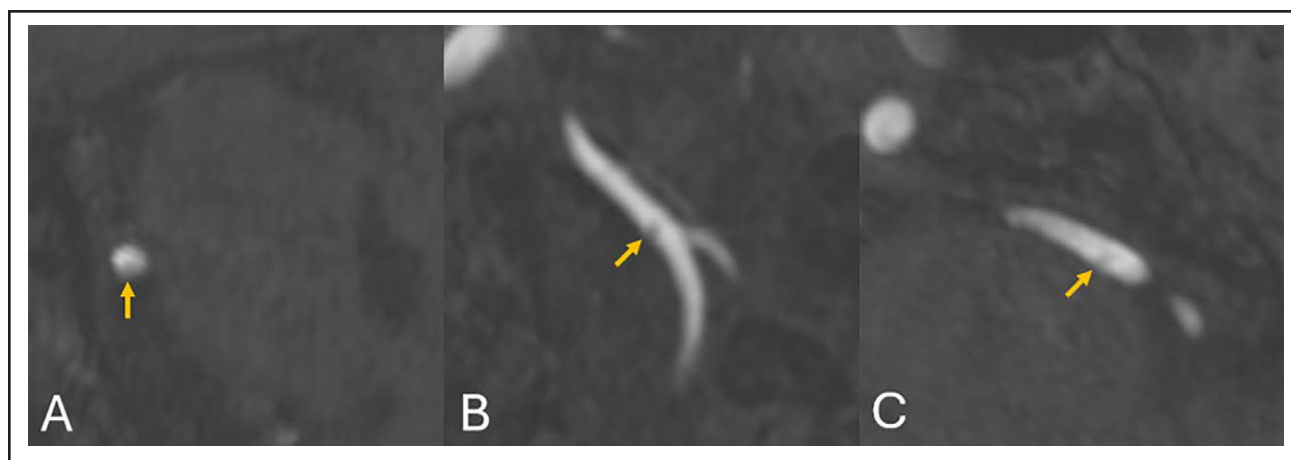


Figure 1. Case 1.

Multiplanar reconstruction of a noncontrast magnetic resonance angiography of the brain acquired with a time-of-flight technique and 0.625-mm isometric voxels. The images of the basilar artery are reconstructed with the axial (**A**) image parallel to the shelf (orange arrow) with the 2 additional perpendicular planes (**B** and **C**).

of atherosclerosis in intracranial or extracranial vessels. The patient was treated symptomatically with intravenous magnesium, metoclopramide, and butalbital, acetaminophen, and caffeine combination, with partial relief of symptoms. He was discharged on his prior regimen of clopidogrel. No further cerebrovascular events have occurred at 2-year follow-up, and no additional imaging has been obtained.

Case 3

A 29-year-old White woman with a medical history notable only for oral contraceptive use initially presented to the emergency department with left-sided paresthesia and seizures. Imaging revealed superior sagittal sinus and right cortical vein thrombosis with venous infarction. She underwent urgent management with heparin, endovascular thrombectomy, and catheter-directed thrombolysis.

Hemorrhagic expansion necessitated decompressive hemicraniectomy with hematoma evacuation. During her hospitalization, she was found to have factor V Leiden thrombophilia. She was discharged on apixaban and antiseizure medications. Retrospective review of her magnetic resonance angiography (no contrast) and DSA from this admission demonstrated a proximal basilar artery web, which was unrelated to her neurological disease (Figure 3A through 3C).

Twenty-one months later, she represented with acute left ptosis and facial weakness. A stroke code was activated; noncontrast head computed tomography showed no hemorrhage, CTA/computed tomography venography excluded large vessel occlusion or recurrent venous thrombosis, and magnetic resonance imaging of the brain showed no acute infarction. CTA confirmed the previously overlooked proximal basilar artery web (Figure 4D

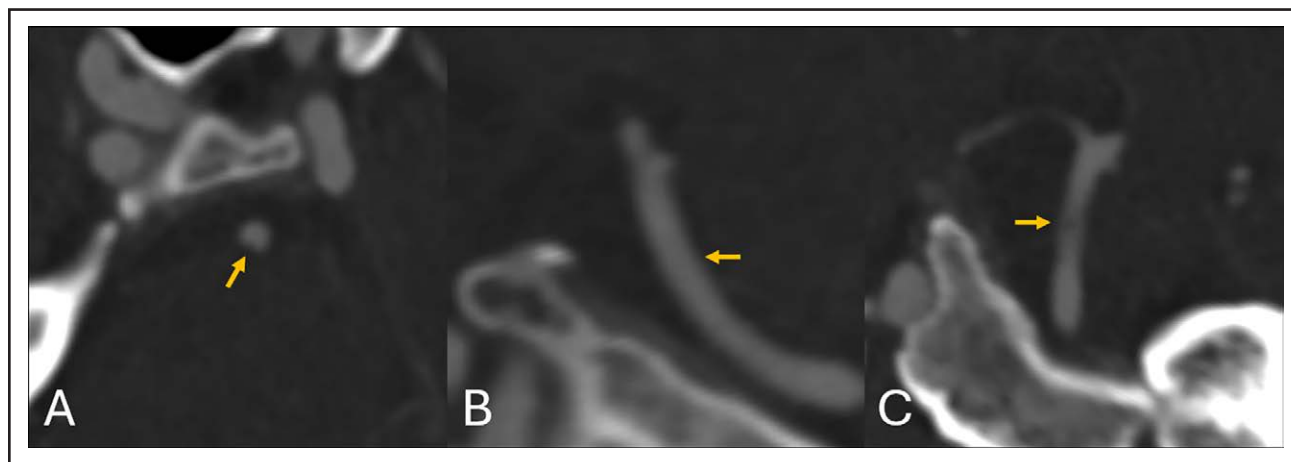


Figure 2. Case 2.

Multiplanar reconstruction of a noncontrast magnetic resonance angiography of the brain acquired with a time-of-flight technique and 0.625-mm isometric voxels. The images of the basilar artery are reconstructed with the axial (**A**) image parallel to the shelf (orange arrow) with the 2 additional perpendicular planes (**B** and **C**).

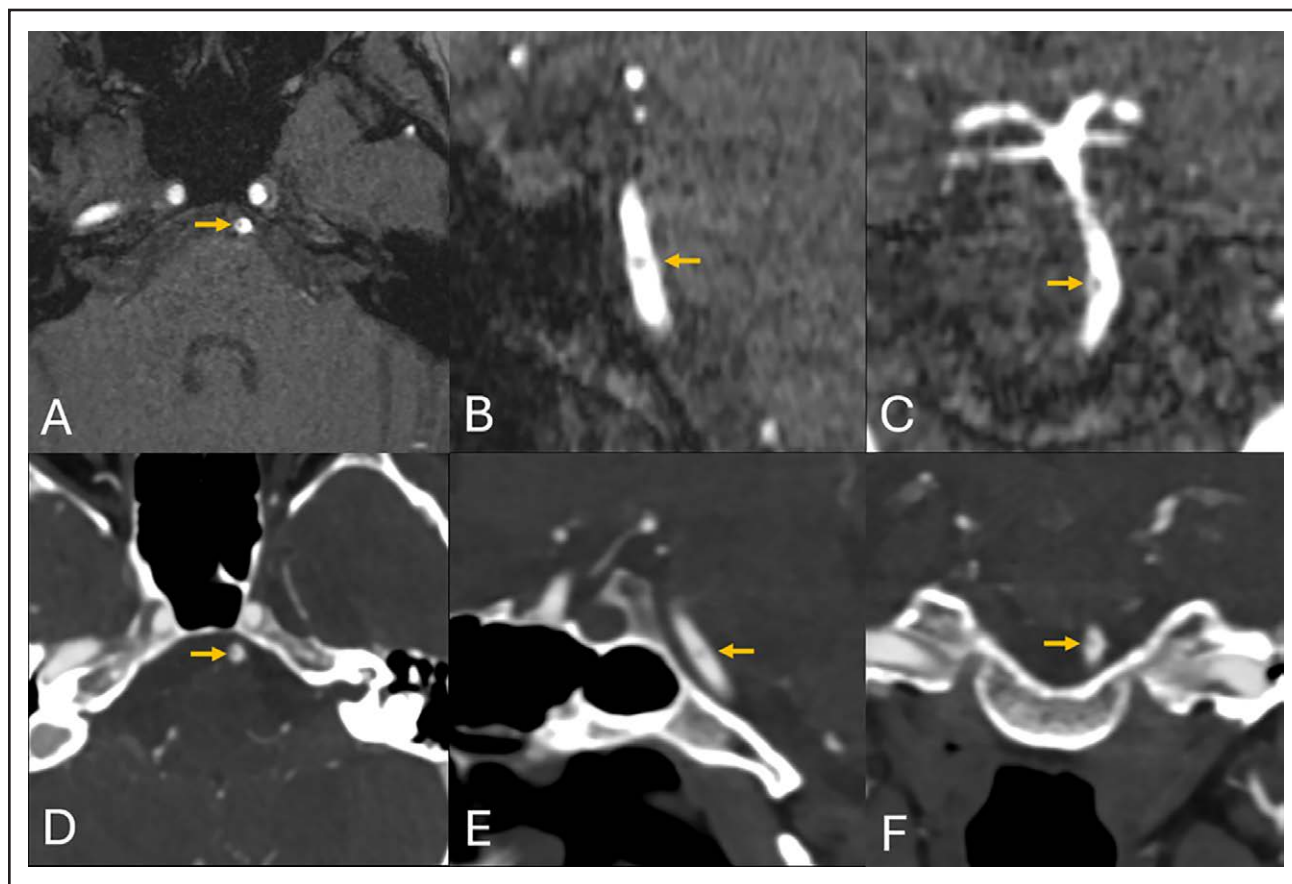


Figure 3. Case 3.

Multiplanar reconstruction of a magnetic resonance angiography (A–C) and computed tomography angiography (D–F) of the brain acquired with 0.625- and 0.7-mm isometric voxels, respectively. The images of the basilar artery are reconstructed with the axial (A and D) images parallel to the shelf with the 2 additional perpendicular planes also shown (B, C, E, and F).

through 4F). Her neurological examination normalized during hospitalization. She was discharged on aspirin 81 mg daily. At 9-month follow-up, she remained without recurrent symptoms.

Case 4

A 45-year-old Hispanic man with hypertension, type 2 diabetes, myocardial infarction treated with coronary stents, and a 45-pack-year former smoking history

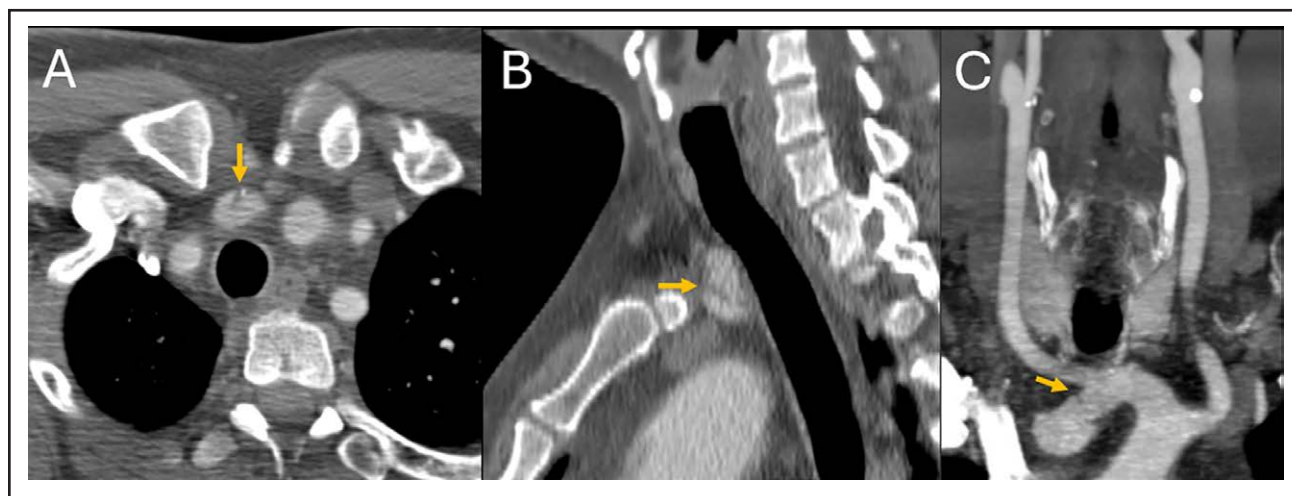


Figure 4. Multiplanar reconstruction of a computed tomography angiography of the brain acquired with 0.7-mm isometric voxels.

The right subclavian artery is reconstructed with the web (orange arrow) on axial (A), sagittal (B), and coronal (C) images.

underwent CTA of the neck as part of an evaluation for possible carotid atherosclerotic disease. He was taking amlodipine, atenolol, valsartan, atorvastatin, and ticagrelor. He denied neurological symptoms.

CTA revealed mild (20%–30%) bilateral carotid bulb atherosclerotic stenosis with eccentric mixed plaques and an incidental linear intraluminal filling defect in the proximal right subclavian artery just distal to the origin of the right common carotid artery (Figure 4A through 4C). The lesion had a smooth surface, lacked calcification, and was morphologically distinct from plaque or focal dissection, consistent with a subclavian artery web. The patient was lost to follow-up.

Systematic Review

Our search query yielded 332 total publications (Preferred Reporting Items for Systematic Reviews and Meta-Analyses flowchart; Figure S2). After screening eligible abstracts, 18 articles underwent full-text review. Of those, 2 articles were excluded.^{12,13} Thus, a total of 16 articles were included in the final review (Table S1), reporting a total of 22 patients.

Including the 4 patients from the present series and 22 patients from 16 published reports, a total of 26 extra-carotid webs were identified. Individual cases are detailed in Table 1, and aggregate characteristics are summarized in Table 2.

The mean age was 52.7 (range, 18–76) years, with 61.5% male. Twelve patients (46.2%) were younger than 60 years of age, and 7 (26.9%) were younger than 40 years of age. The vertebral artery was the most common extra-carotid web location (53.8%). Approximately one-third of patients were symptomatic, defined as having an ipsilateral stroke or transient ischemic attack without a more likely competing stroke cause. Excluding our institutional case series (none of our patients had symptomatic webs), 4 of 5 basilar webs and 6 of 14 vertebral webs were symptomatic. Of those 10 symptomatic posterior circulation webs, 4 of 10 were associated with recurrent strokes. No patients with subclavian webs were symptomatic.

DSA was the primary imaging modality for webs (42.3%), with multimodal imaging applied in 5 cases. On DSA, webs appeared as smooth, focal lesions with a shelf-like ledge, causing moderate to mild stenosis without flow limitation but with evidence of focal flow turbulence or eddy current. In some cases, contrast stagnation was also noted in the ledge late into the venous phase. Specific imaging modalities varied on a case-by-case basis. Antiplatelet/antithrombotic therapy (53.8%) or observation (34.6%) was the most frequent treatment approach.

Endovascular stenting was reserved for 3 patients (11.5%) with V1, V3, and ostial vertebral webs who experienced recurrent cerebellar and occipital lobe strokes despite antiplatelet therapy. Notably, all patients who underwent stenting had no recurrent symptoms. The only

reported stroke recurrence after treatment occurred in a patient with a basilar web who discontinued anticoagulation, developing a late ischemic event a decade later at the age of 28 years. Follow-up data were available for only 6 patients across the literature.

One patient with a vertebral web reported in the literature was noted to have calcifications next to the vertebral web and had atherosclerotic plaques in the bilateral subclavian arteries. This case was retained to accurately reflect the reported literature, despite the authors' notation of concurrent atherosclerosis and calcification, which raises the possibility of an atherosclerotic plaque mimicking a web. However, a definitive diagnosis remains uncertain due to the lack of pathological confirmation.²⁵

DISCUSSION

Combining 4 cases of extra-carotid arterial webs with prior literature, we present 26 patients with webs involving the vertebral, basilar, or subclavian arteries. Although rare, these arterial lesions share clinical and radiographic features with carotid webs, and vertebral and basilar webs may contribute to ischemic stroke via a similar thromboembolic mechanism.

Basilar Webs

In our cohort, all basilar webs were asymptomatic, contrasting with most published cases, in which these lesions were frequently associated with recurrent posterior circulation strokes in otherwise young, healthy individuals; the oldest reported patient was 51 years of age.^{6,8,9,16,22} This discrepancy likely reflects publication bias (much of the current literature consists of case reports, whereas our case series resulted from a systematic review of our institution's angiography findings).

However, it also illustrates the need to clarify which basilar webs carry true embolic potential versus which are incidental imaging findings. Such risk stratification work has been explored with carotid webs and could feasibly be extended to extra-carotid arterial webs.^{4,26} Unlike carotid webs, which cluster at the carotid bulb, basilar webs have been described at varied sites along the artery.^{8,9,16,22} There might be site-specific hemodynamic effects, raising questions regarding whether webs in certain segments have a greater predisposition toward thromboembolism. Careful differentiation from fenestrations or atherosclerotic plaque remains essential, as these mimics are far more common.^{13,27}

Vertebral Webs

Our institutional review demonstrates how vertebral artery tortuosity and fenestrations can often take on a shelf-like appearance and act as web mimics, emphasizing the importance of rigorous multimodality imaging.²⁸ Nonetheless, vertebral webs remain the most

Table 1. Summary of Reported Extra-Carotid Arterial Webs

Study	Age, y; sex	Vascular risk factors	Web location	Clinical presentation	Imaging modality	Treatment	Outcome
Green and Caplan, 2000 ⁶	18, M	None	Basilar	Pontine stroke	DSA	Warfarin, discontinued after several months	Stroke recurrence 10 y later, managed with aspirin, no recurrence since
Lenck et al, 2014 ¹⁴	23, M	None	Vertebral (V3)	Recurrent cerebellar, thalamic, and occipital stroke	DSA	Stent	No recurrence at 1-y follow-up
	57, M*	4 cardiovascular risk factors, unspecified	Vertebral (ostium)	Recurrent cerebellar/occipital strokes	CTA	Stent	No recurrence at 6-mo follow-up
	69, M	2 cardiovascular risk factors, unspecified	Vertebral (V3)	Recurrent cerebellar/occipital strokes	DSA	Dual-antiplatelet therapy	No recurrence at 18-mo follow-up
Pereira et al, 2018 ¹⁵	66, F	None	Vertebral	Incidental	DSA	Observation only	Not reported
	69, M	None	Vertebral and concurrent carotid web	Incidental and anterior circulation stroke from the carotid web	DSA	Dual-antiplatelet therapy after stenting for the carotid web	Not reported
	75, F	None	Vertebral (V1)	Recurrent cerebellar strokes	DSA	Stent	Not reported
Chen et al, 2019 ¹⁰	50, F	None	Vertebral (ostium)	Recurrent anterior circulation TIAs	DSA and ultrasound	Dual-antiplatelet therapy	No recurrence, timeframe unspecified
Esenwa et al, 2019 ¹⁶	19, F	None	Basilar	Pontine stroke	MRA	Antiplatelet (aspirin)	Stable at 11-mo follow-up
Dargazanli et al, 2019 ¹⁷	71, M	None	Vertebral (V2)	Cerebellar and posterior lobe stroke	CTA	Antiplatelet (aspirin)	Stable at 3-mo follow-up
Wang et al, 2021 ⁷	56, M	HTN	Vertebral (transverse foramen)	Incidental and SAH (basilar aneurysm)	CTA	Not reported	Not reported
	64, F	None	Subclavian	Incidental	CTA	Not reported	Not reported
	74, M	HTN, HLD, and DM2	Subclavian	Incidental (headache and dizziness workup)	CTA	Not reported	Not reported
Wang et al, 2022 ¹⁸	67, F	HTN	Vertebral (V4) and concurrent carotid web	Incidental (vertigo workup)	MRA	Antiplatelet (aspirin)	Not reported
Robb et al, 2022 ¹⁹	75, M†	Classic FMD and aortic dissection, otherwise unspecified	Subclavian	Incidental and FMD-associated	CTA	Observation	Not reported
Li and Chen, 2023 ⁹	34, M	None	Basilar	Pontine stroke	DSA and MRA	Antiplatelet (aspirin)	Not reported
Blanco-Ruiz et al, 2023 ²⁰	60, F	None	Vertebral (V1)	Incidental and M2 stroke due to a carotid web	CTA	Not reported	Not reported
Waqar et al, 2024 ²¹	31, F	None	Vertebral (V3)	Recurrent cerebellar, pontine, and thalamic stroke	DSA	Antiplatelet (aspirin), stent considered if recurrent	Not reported
Fernandez-Vidal et al, 2024 ²²	48, M	HTN, HLD, DM2, and infrequent cocaine use	Basilar	Occipital stroke	CTA and HR-MRI/MRA	Antiplatelet (aspirin)	Not reported
Xu et al, 2025 ⁹	51, M	None	Basilar (dual webs)	Incidental and involuntary movements with no infarct	DSA and HR-MRI/MRA	Not reported	Not reported
Li et al, 2025 ²³	67, F	HTN	Vertebral (V4)	Incidental (vertigo workup)	HR-MRI/MRA	Not reported	Not reported
Huang et al, 2025 ²⁴	45, M	None	Vertebral (V4)	Incidental (headache workup)	HR-MRI/MRA	Warfarin	Stable at 3-mo follow-up

(Continued)

Table 1. Continued

Study	Age, y; sex	Vascular risk factors	Web location	Clinical presentation	Imaging modality	Treatment	Outcome
Institutional case series	76, M	HLD (well-controlled)	Basilar	Incidental (dizziness workup)	HR-MRI/MRA	Antiplatelet (aspirin, self-discontinued)	Stable at 4-y follow-up
	31, M	2.5-pack-year smoking history (quit) and prior stroke from PFO (since closed)	Basilar	Incidental (headache workup)	MRA	Antiplatelet (clopidogrel)	Stable at 2-y follow-up
	29, F	None and prior CVST due to factor V Leiden	Basilar	Incidental (CVST workup)	CTA, MRA, and DSA	Antiplatelet (aspirin)	Stable at 9-mo follow-up
	45, M	HTN, HLD, DM2, prior MI with coronary stents, and 45-pack-year smoking history (quit)	Subclavian	Incidental (carotid stenosis workup)	CTA	Observation	Lost to follow-up

CTA indicates computed tomography angiography; CVST, cerebral venous sinus thrombosis; DM2, diabetes type 2; DSA, digital subtraction angiography; F, female; FMD, fibromuscular dysplasia; HLD, hyperlipidemia; HR, high-resolution; HTN, hypertension; M, male; MI, myocardial infarction; MRA, magnetic angiography; MRI, magnetic resonance imaging; PFO, patent foramen ovale; SAH, subarachnoid hemorrhage; and TIA, transient ischemic attack.

*This patient's web was noted to be associated with atherosclerotic change, with calcifications noted next to the web.

†This patient's subclavian web was discovered in association with classic medial FMD.

frequently reported extra-carotid arterial webs, with 13 cases described across the V1, V3, V4, and ostial segments and over half linked to posterior circulation strokes.^{10,14,15,20,21,24} In 2 cases, vertebral webs were found in patients who also harbored carotid webs, suggesting the potential for multifocal disease (as has been previously observed in patients with bilateral carotid webs).^{7,15,18,29}

Importantly, the few reported cases of recurrent stroke despite antiplatelet therapy responded well to vertebral artery stenting, suggesting that, as with carotid webs, medical therapy alone may not provide sufficient secondary prevention for high-risk patients.^{14,15,30}

Subclavian Webs

Subclavian webs remain rare, with more common pathologies such as short-segment dissection, atherosclerotic plaque, or imaging artifact comprising more common mimics. In the literature, the subclavian artery is the least commonly reported arterial web site, with only 2 cases described as isolated lesions and 1 case described in the context of classic FMD, raising the question of whether that case represented a true web or a broader arteriopathy.^{7,19} Unlike carotid, vertebral, or basilar webs, subclavian webs lack a clear link to cerebral embolism. Nonetheless, their recognition expands the anatomic spectrum of webs and supports the concept that webs can occur throughout the cervicothoracic arterial tree.⁷

Toward a Unified Concept of Cervical Artery Webs

The consistent shelf-like intraluminal morphology across carotid and extra-carotid sites supports the possibility of a single disease spectrum, cervical artery web, as proposed by Wang et al,⁷ encompassing webs of the carotid, vertebral, basilar, and subclavian arteries, with carotid webs being the most common. Histologically, carotid webs are characterized by focal eccentric intimal hyperplasia, distinct from the medial dysplasia characteristic of

classic FMD.^{2,5,31} Thus, while carotid webs were initially considered an atypical variant of FMD, accumulating evidence suggests that they constitute a distinct entity.³² This distinction is likely relevant to extra-carotid webs if they share a similar pathology.

Morphological variability further supports a spectrum of disease. Carotid webs, although classically described along the posterior wall of the carotid bulb, have been documented with lateral, medial, and anterior attachments.^{2,5,7} Similarly, basilar and vertebral webs occur at variable segmental locations (eg, V1/V3/V4 and variable basilar segments).^{9,14,15}

Notably, the underlying disease cause remains unclear. Early hypotheses proposed carotid webs as developmental anomalies of the brachiocephalic arteries, while later work has suggested contributions from genetic predisposition, chronic vascular injury, or hormonal influences.^{1,33,34} Defining the underlying pathogenesis is an important direction for future research.

Conceptualizing webs as a unified entity carries both clinical and scientific implications. Clinically, it broadens the differential for cryptogenic posterior circulation stroke, emphasizing the importance of targeted vascular imaging and consideration of intervention when posterior circulation ischemic events recur despite antiplatelet therapy. Scientifically, it provides a framework for future research into whether cervical webs are congenital or acquired, the potential role of genetic or environmental drivers, and the selective predilection for certain arterial segments, such as the carotid bulb.

Imaging Considerations

DSA remains the most reliable modality for detecting subtle intraluminal webs and should be considered when less invasive studies are inconclusive.^{1,4} CTA is widely available and can help differentiate webs from atherosclerotic plaque, as webs lack circumferential wall thickening or calcification.^{1,25,35,36}

Table 2. Aggregate Characteristics of Patients With Extra-Carotid Arterial Webs

	n (%)
Total patients	26
Systematic review	22 (84.6)
Present series	4 (15.4)
Age, y	
Median	56.5
Range	18–76
Sex	
Male	16 (61.5)
Female	10 (38.5)
Web vessel	
Vertebral	14 (53.8)
Basilar	8 (30.8)
Subclavian	4 (15.4)
No. of cardiovascular risk factors	
0	15 (57.7)
1	5 (19.2)
2	1 (3.8)
3	2 (7.7)
4	2 (7.7)
Clinical presentation	
Symptomatic	10 (38.5)
Asymptomatic	16 (61.5)
Imaging modalities for diagnosis	
DSA	11 (42.3)
CTA	10 (38.5)
MRI/MRA	10 (38.5)
Ultrasound	1 (3.8)
Multimodal*	5 (19.2)
Treatment	
Medical management	14 (53.8)
Endovascular stenting	3 (11.5)
Observation/no treatment/not reported	9 (34.6)
Follow-up (n=10), mo	
Median	10
Range	3–120

CTA indicates computed tomography angiography; DSA, digital subtraction angiography; MRA, magnetic resonance angiography; and MRI, magnetic resonance imaging.

*Including CTA, MRA, or DSA in different combinations.

High-resolution vessel-wall magnetic resonance imaging offers additional specificity by demonstrating smooth, focal thickening.^{23,37,38} In our experience, carotid webs may also show postcontrast T1 enhancement, a potentially useful supportive feature that we also observed in one of the patients in our series with a basilar web. Multiplanar, multimodality evaluation is critical to exclude web mimics such as fenestration, dissection, vessel tortuosity, or plaque before diagnosing a web.

Limitations

A significant limitation of this study is the likely underrepresentation of extra-carotid arterial webs in the reviewed literature. This scarcity is rooted in the limited clinical awareness of the condition. As carotid webs have only recently gained recognition and remain underdiagnosed, extra-carotid webs are likely even more common than currently appreciated. Another important factor is publication bias: symptomatic webs associated with stroke are far more likely to be reported, whereas incidental or asymptomatic cases, such as those identified in our series, often go unrecognized. This pattern mirrors that seen with carotid webs, where research disproportionately emphasizes symptomatic cases.

However, this study's primary limitation is arguably the lack of ex vivo histopathologic visualization and, thus, a lack of diagnostic certainty. While carotid webs are often diagnosed radiographically in practice, definitive diagnosis requires endarterectomy specimens, which are unavailable for extra-carotid sites in the literature.⁵ In our series, we performed meticulous imaging review to minimize misclassification, but the absence of tissue validation remains a notable constraint, as only histopathologic analysis can definitively differentiate a web from nonocclusive thrombus, dissection, or other pathologies. When histopathologic analysis is unavailable, DSA with assessment of contrast pooling may augment diagnostic sensitivity, and thus, cases lacking DSA are also limited in this respect. Furthermore, available evidence is limited to isolated case reports and small case series with heterogeneous imaging protocols and limited follow-up, restricting conclusions about natural history and treatment outcomes.

Conclusions

Extra-carotid webs, comprising arterial webs of the basilar, vertebral, and subclavian arteries, appear to share morphology and clinical behavior with carotid webs, suggesting a unified disease spectrum of cervical arterial webs. Basilar and vertebral webs may also underlie a subset of otherwise cryptogenic posterior circulation strokes. Recognizing cervical artery webs as a unified entity may improve diagnostic precision, align management with recurrence risk, and guide further research in clarifying these lesions' natural history, optimal management strategies, and underlying cause. However, the lack of ex vivo histopathologic visualization remains a major limitation of our study and, thus, represents an important area of future investigation.

ARTICLE INFORMATION

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Affiliations

Department of Neurosurgery (E.A.G., E.N.), Department of Radiology (E.G.S.), and Department of Neurology (L.B., A.C., S.K.R., M.R.), NYU Grossman School of Medicine, New York, NY. Department of Neurology, NYC Health+Hospitals/

Bellevue, New York, NY (L.B., S.K.R., M.R.). Department of Neurology, University of Pennsylvania, Philadelphia (S.E.K.). Neuro Clinical, Bayer U.S. LLC, Whippany, NJ (S.E.K.).

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Disclosures

Dr Kasner is an employee of Bayer. Dr Nossek is a consultant for Rapid Medical and Longeviti Neuro Solutions, serves on the medical board of Longeviti, and holds stock equity or options in Rapid Medical and Longeviti. The other authors report no conflicts.

Supplemental Material

Table S1

Figures S1–S2

REFERENCES

- Chen H, Colasurdo M, Costa M, Nossek E, Kan P. Carotid webs: a review of pathophysiology, diagnostic findings, and treatment options. *J Neurointerv Surg*. 2024;16:1294–1298. doi: 10.1136/jnis-2023-021243
- Grin EA, Wiggan DD, Rosso M, Sharashidze V, Chung C, Stein E, Shapiro M, Raz E, Baranoski J, Riina HA, et al. Carotid webs [published online November 26, 2025]. *Semin Neurol*. 2025. doi: 10.1055/a-2741-2575
- Compagne KCJ, Dilba K, Postema EJ, van Es A, Emmer BJ, Majoie C, van Zwam WH, Dippel DWJ, Wentzel JJ, van der Lugt A, et al; for the MR CLEAN Investigators. Flow patterns in carotid webs: a patient-based computational fluid dynamics study. *AJNR Am J Neuroradiol*. 2019;40:703–708. doi: 10.3174/ajnr.a6012
- Negash B, Wiggan DD, Grin EA, Sangwon KL, Chung C, Gutstadt E, Sharashidze V, Raz E, Shapiro M, Ishida K, et al. Use of carotid web angioarchitecture in stratification of stroke risk [published online June 20, 2025]. *J Neurointerv Surg*. 2025. doi: 10.1136/jnis-2025-023368
- Grin EA, Raz E, Shapiro M, Sharashidze V, Negash B, Wiggan DD, Belakhova S, Sangwon KL, Ishida K, Torres J, et al. Atypical carotid webs: an elusive etiology of ischemic stroke. *World Neurosurg*. 2025;196:123770. doi: 10.1016/j.wneu.2025.123770
- Green DM, Caplan LR. Recurrent basilar branch infarcts due to a protruding basilar artery lesion. *Cerebrovasc Dis*. 2000;10:489. doi: 10.1159/000016118
- Wang M, Zhou R, Zhao H, Niu L, Liu S, Li Y, Liu X. Imaging and clinical features of cervical artery web: report of 41 cases and literature review. *Acta Neurol Belg*. 2021;121:1225–1233. doi: 10.1007/s13760-020-01353-y
- Li J, Chen L. Brainstem infarction due to a basilar arterial web. *Radiology*. 2023;309:e231106. doi: 10.1148/radiol.231106
- Xu Z, Shi Y, Yang S, Li Y. A rare case of bilateral basilar artery webs. *J Stroke Cerebrovasc Dis*. 2025;34:108393. doi: 10.1016/j.jstrokecerebrovasdis.2025.108393
- Chen L, Du L, Wen X, Jiang Y, Hu F, Zheng H. Is artery web seen in vertebral artery? *J Stroke Cerebrovasc Dis*. 2019;28:104440. doi: 10.1016/j.jstrokecerebrovasdis.2019.104440
- Jiang LY, Liu XC, Nejatian NP, Nasir-Moin M, Wang D, Abidin A, Eaton K, Riina HA, Laufer I, Punjabi P, et al. Health system-scale language models are all-purpose prediction engines. *Nature*. 2023;619:357–362. doi: 10.1038/s41586-023-06160-y
- Lipchik EO, DeWeese JA, Schenk EA, Lim GH. Diaphragm-like obstructions of the human arterial tree. *Radiology*. 1974;113:43–46. doi: 10.1148/113.1.43
- Slugocki M, Rzeplinski R, Tarka S, Stepień T, Tomaszewski M, Kuciewicz M, Kępczyńska G, Małachowski J, Ciszek B. Endovascular structures of the basilar artery as forms of the basilar nonfusion spectrum. *Sci Rep*. 2025;15:28192. doi: 10.1038/s41598-025-14558-z
- Lenck S, Labeyrie MA, Saint-Maurice JP, Tarlov N, Houdart E. Diaphragms of the carotid and vertebral arteries: an under-diagnosed cause of ischaemic stroke. *Eur J Neurol*. 2014;21:586–593. doi: 10.1111/ene.12343
- Pereira BJA, Batista UC, Tosello RT, Stroher IN, Baeta AM, Piske RL. Web vessels: literature review and neurointerventional management. *World Neurosurg*. 2018;110:e907–e916. doi: 10.1016/j.wneu.2017.11.115
- Esenwa C, Labovitz D, Caplan LR. "Basilar web" causing basilar branch infarction. *J Stroke Cerebrovasc Dis*. 2019;28:104366. doi: 10.1016/j.jstrokecerebrovasdis.2019.104366
- Dargazanli C, Arnaud I, Seiller I, Signate A, Mejoudi M. Successful mechanical thrombectomy of acute basilar artery occlusion caused by vertebral artery intimal dysplasia (web). *Rev Neurol (Paris)*. 2020;176:210–213. doi: 10.1016/j.neuro.2019.05.003
- Wang Z, Mao Y, Zhao H, Ling Y, Cheng X, Cao W. Carotid web coexisting with vertebral web in a middle-aged woman. *Neurol India*. 2022;70:826–827. doi: 10.4103/0028-3886.344632
- Robb CL, Bhalla S, Raptis CA. Subclavian artery: anatomic review and imaging evaluation of abnormalities. *Radiographics*. 2022;42:2149–2165. doi: 10.1148/rg.220064
- Blanco-Ruiz M, Pinilla RM, Perez JF, Gomez MF, Padillo AA. Are cervical webs found only in the carotid arteries? A case of vertebral web. *Neurol India*. 2023;71:779–780. doi: 10.4103/0028-3886.383836
- Waqar K, Kesav P, John S. Vertebral web and embolic stroke. *JAMA Neurol*. 2025;82:104–105. doi: 10.1001/jamaneurol.2024.4025
- Fernandez-Vidal JM, Guasch-Jimenez M, Ruiz-Barrio I, Gomez-Anson B, Tecame M, Marti-Fabregas J. Basilar web and basilar fenestration: a case report. *Neurologia (Engl Ed)*. 2024;39:209–210. doi: 10.1016/j.nrleng.2024.01.010
- Li J, Peng F, Zhao X. Teaching neuroimage: vessel wall imaging of a vertebral artery web. *Neurology*. 2025;104:e210175. doi: 10.1212/WNL.0000000000210175
- Huang W, Xu K, Jin H. Vessel wall imaging of a possible intracranial vertebral artery web. *Stroke*. 2025;56:e238–e239. doi: 10.1161/STROKEAHA.125.052707
- Grin EA, Chen A, Koneru S, Sharashidze V, Schneider JR, Ayoub G, Raz E, Shapiro M, Rostanski SK, Nossek E, et al. Misleading morphology: histological diagnosis reveals atherosclerotic plaque mimicking carotid web. *Stroke Vasc Interv Neurol*. 2025;6:e002103. doi: 10.1161/svin.125.002103
- von Oiste GG, Sangwon KL, Chung C, Narayan V, Raz E, Shapiro M, Rutledge C, Nelson PK, Ishida K, Torres JL, et al. Use of carotid web angioarchitecture for stroke risk assessment. *World Neurosurg*. 2024;182:e245–e252. doi: 10.1016/j.wneu.2023.11.091
- Abdelhamid HM, Bhatt NR, Viana LS, Ferreira FM, Nogueira RG, Al-Bayati AR, Grossberg JA, Allen JW, Haussen DC. Multiplane reconstruction modifies the diagnostic performance of CT angiography in carotid webs. *Clin Neurol Neurosurg*. 2024;244:108441. doi: 10.1016/j.clineuro.2024.108441
- Bitar A, Almahder D, Jouini JA, Alsaïd B. Cervical arteries tortuosity and its association with dissection: a systematic review and meta-analysis. *Medicine (Baltim)*. 2025;104:e41517. doi: 10.1097/MD.00000000000041517
- Essibayi MA, Nasr D, Lanzino G. Bilateral carotid webs. *Neuroradiol J*. 2021;34:683–685. doi: 10.1177/19714009211017783
- Bounajem MT, Liang A, Trang A, El Baba B, Bielinski TM, Sangwon K, Zhang Y, Wiggan D, Grin E, Gajjar A, et al. Outcomes after carotid revascularization for symptomatic carotid artery web: a multi-institutional cohort study [published August 8, 2025]. *Interv Neurol*. 2025;15910199251365529. doi: 10.1177/15910199251365529
- Wang LZ, Calvet D, Julia P, Domigo V, Mohamedi N, Alsac JM, El Batti S, Messas E, Mirault T, Bruneval P, et al. Is carotid web an arterial wall dysplasia? A histological series. *Cardiovasc Pathol*. 2023;66:107544. doi: 10.1016/j.carpath.2023.107544
- Sharashidze V, Nogueira RG, Al-Bayati AR, Bhatt N, Nahab FB, Yun J, Allen JW, Frankel M, Haussen DC. Carotid web phenotype is uncommonly associated with classic fibromuscular dysplasia: a retrospective observational study. *Stroke*. 2022;53:e33–e36. doi: 10.1161/STROKEAHA.121.036188
- McNamara MF. The carotid web: a developmental anomaly of the brachiocephalic system. *Ann Vasc Surg*. 1987;1:595–597. doi: 10.1016/S0890-5096(06)61448-9
- Singh RJ, Mandell DM, Menon BK, Appireddy R. De novo formation of a carotid web in an adult: a longitudinal observation. *J Stroke Cerebrovasc Dis*. 2024;33:108010. doi: 10.1016/j.jstrokecerebrovasdis.2024.108010
- Choi PM, Singh D, Trivedi A, Qazi E, George D, Wong J, Demchuk AM, Goyal M, Hill MD, Menon BK. Carotid webs and recurrent ischemic strokes in the era of CT angiography. *AJNR Am J Neuroradiol*. 2015;36:2134–2139. doi: 10.3174/ajnr.A4431
- Mac Grory B, Emmer BJ, Roosendaal SD, Zagzag D, Yaghi S, Nossek E. Carotid web: an occult mechanism of embolic stroke. *J Neurol Neurosurg Psychiatry*. 2020;91:1283–1289. doi: 10.1136/jnnp-2020-323938
- Madaelil TP, Grossberg JA, Nogueira RG, Anderson A, Barreira C, Frankel M, Haussen DC. Multimodality imaging in carotid web. *Front Neurol*. 2019;10:220. doi: 10.3389/fneur.2019.00220
- Saba L, Yuan C, Hatsukami TS, Balu N, Qiao Y, DeMarco JK, Saam T, Moody AR, Li D, Matouk CC, et al; Vessel Wall Imaging Study Group of the American Society of Neuroradiology. Carotid artery wall imaging: perspective and guidelines from the ASNR vessel wall imaging study group and expert consensus recommendations of the American Society of Neuroradiology. *AJNR Am J Neuroradiol*. 2018;39:E9–E31. doi: 10.3174/ajnr.A5488